JONGHA (JON) RYU

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RESEARCH INTERESTS

My research develops spectral and information-theoretic foundations for machine learning, with an emphasis on scalable algorithms grounded in mathematical principles. Drawing on my training in information theory, I seek to identify unifying principles across seemingly disparate problems and translate them into practical learning methods. Recent work includes neural network-based operator learning for scientific simulation [C9], [C15], [C16], score-based generative modeling [C12], and uncertainty quantification and anytime-valid inference [J2], [C10], [C11], [C14].

EMPLOYMENT

Pennsylvania State University Aug. 2026 – present
Wormley Family Early Career Assistant Professor, Department of Computer Science and Engineering
· Faculty co-hire with the Institute for Computational and Data Sciences (ICDS)

Massachusetts Institute of Technology (MIT) Aug. 2022 – Aug. 2026
Postdoctoral Associate, Research Laboratory of Electronics (RLE)
· Advisor: Prof. Gregory W. Wornell

EDUCATION

University of California San Diego (UCSD) Sep. 2015 – Jun. 2022
Ph.D. in Electrical Engineering (GPA: 3.99/4.0)
· Thesis advisors: Prof. Young-Han Kim and Prof. Sanjoy Dasgupta
· Thesis title: "From Information Theory to Machine Learning Algorithms: A Few Vignettes"

M.S. in Electrical Engineering Dec. 2018

Seoul National University (SNU) Mar. 2008 – Aug. 2015
Bachelor of Science (*summa cum laude*, GPA: 4.11/4.3)
· Double major in **Electrical and Computer Engineering** & **Mathematical Sciences**; minor in **Physics**

JOURNAL PAPERS

(* indicates equal contribution. † indicates alphabetical orders.)

[J1] **J. Jon Ryu***, Shouvik Ganguly*, Young-Han Kim, Yung-Kyun Noh, Daniel Lee, "Nearest neighbor density functional estimation from inverse Laplace transform". *IEEE Trans. Info. Theory*, vol. 68, no. 6, pp. 3511-3551, Jun. 2022.

[J2] **J. Jon Ryu**, Alankrita Bhatt, "On Confidence Sequences for Bounded Random Processes via Universal Gambling Strategies," *IEEE Trans. Info. Theory*, vol. 70, no. 10, pp. 7143-7161, Oct. 2024.

CONFERENCE PAPERS

[C1] Alankrita Bhatt†, Jiun-Ting Huang†, Young-Han Kim†, **J. Jon Ryu**†, Pinar Sen†, "Monte Carlo methods for randomized likelihood decoding," *Annual Allerton Conference on Communication, Control, and Computation (Allerton)*, Sep. 2018.

[C2] **Jongha Ryu**, Young-Han Kim, "Conditional distribution learning using neural networks and its application to universal image denoising," *IEEE International Conference on Image Processing (ICIP)*, Oct. 2018.

[C3] Alankrita Bhatt†, Jiun-Ting Huang†, Young-Han Kim†, **J. Jon Ryu**†, Pinar Sen†, "Variations on a theme by Liu, Cuff, and Verdú: The power of posterior sampling," *IEEE Information Theory Workshop (ITW)*, Nov. 2018.

[C4] Yang Yang, Guillaume Sautiere, **J. Jon Ryu**, Taco Cohen, "Feedback Recurrent Autoencoder," *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2020.

[C5] **J. Jon Ryu**, Jiun-Ting Huang, Young-Han Kim, "On the Role of Eigendecomposition in Kernel Embedding," *IEEE International Symposium on Information Theory (ISIT)*, Jun. 2021.

[C6] **J. Jon Ryu**, Alankrita Bhatt, Young-Han Kim, "Parameter-free Online Linear Optimization with Side Information via Universal Coin Betting," *International Conference on Artificial Intelligence and Statistics (AISTATS)*, Apr. 2022.

[C7] Alankrita Bhatt*, **J. Jon Ryu***, Young-Han Kim, "On Universal Portfolios with Continuous Side Information," *AISTATS*, Apr. 2023.

[C8] Abhin Shah, Maohao Shen, **J. Jon Ryu**, Subhro Das, Prasanna Sattigeri, Yuheng Bu, Gregory W. Wornell, "Group Fairness with Uncertainty in Sensitive Attributes," *ISIT*, Jul. 2024. (Note: A preliminary version of this manuscript was presented at *ICML 2023 Workshop on Spurious Correlations, Invariance, and Stability*.)

[C9] **J. Jon Ryu**, Xiangxiang Xu, Hasan Sabri Melihcan Erol, Yuheng Bu, Lizhong Zheng, Gregory Wornell, "Operator SVD with Neural Networks via Nested Low-Rank Approximation," *International Conference on Machine Learning (ICML)*, Jul. 2024.

[C10] **J. Jon Ryu**, Gregory W. Wornell, "Gambling-Based Confidence Sequences for Bounded Random Vectors," *ICML*, Jul. 2024. **Spotlight** (top 3.5%).

[C11] Maohao Shen*, **J. Jon Ryu***, Soumya Ghosh, Yuheng Bu, Prasanna Sattigeri, Subhro Das, Gregory W. Wornell, "Are Uncertainty Quantification Capabilities of Evidential Deep Learning a Mirage?," *Annual Conference on Neural Information Processing Systems (NeurIPS)*, Dec. 2024.

[C12] Tejas Jayashankar*, **J. Jon Ryu***, Gregory W. Wornell, "Score-of-Mixture Training: Training One-Step Generative Models Made Simple via Score Estimation of Mixture Distributions," *ICML*, Jul. 2025. **Spotlight** (top 2.6%).

[C13] **J. Jon Ryu**, Abhin Shah, Gregory W. Wornell, "A Unified View on Learning Unnormalized Distributions via Noise-Contrastive Estimation," *ICML*, Jul. 2025.

[C14] **J. Jon Ryu**, Jeongyeol Kwon, Benjamin Koppe, Kwang-Sung Jun, "Improved Offline Contextual Bandits with Second-Order Bounds: Betting and Freezing," *Conference on Learning Theory (COLT)*, Jun. 2025.

[C15] Minchan Jeong*, **J. Jon Ryu***, Se-Young Yun, Gregory W. Wornell, "Efficient Parametric SVD of Koopman Operator for Stochastic Dynamical Systems," *NeurIPS*, Dec. 2025.

[C16] **J. Jon Ryu**, Samuel Zhou, Gregory W. Wornell, "Revisiting Orbital Minimization for Neural Operator Decomposition," *NeurIPS*, Dec. 2025.

[C17] **J. Jon Ryu**, Pavan Yeddanapudi, Xiangxiang Xu, Gregory W. Wornell, "Contrastive Predictive Coding Done Right for Mutual Information Estimation," *International Conference on Learning Representations (ICLR)*, Apr. 2026.

[C18] Hasan Sabri Melihcan Erol, Suat Evren, Oktay Ozel, Alexander Morgan, **J. Jon Ryu**, Lizhong Zheng, "When Softmax Fails at the Top: Extreme-Value Corrections for InfoNCE," *ICML*, Jul. 2026.

[C19] Hasan Sabri Melihcan Erol, **J. Jon Ryu**, Lizhong Zheng, "Beyond Linsker's Infomax Principle: A Rate-Distortion Perspective on InfoNCE," *ITW*, Nov. 2026.

[C20] Minchan Jeong*, **J. Jon Ryu***, Se-Young Yun, Gregory W. Wornell, "Variational Monte Carlo for Quantum Excited States via Nested Low-Rank Approximation," *NeurIPS*, Dec. 2026.

[C21] Jihun Yun, Sungjoon Yoon, Beomhan Baek, Minhak Song, **J. Jon Ryu**, Kwang-Sung Jun, "Beyond Clipping: Signed Logarithmic Smoothing for Policy Optimization," *NeurIPS*, Dec. 2026.

Selected Preprints

[P1] Tejas Jayashankar*, **J. Jon Ryu***, Xiangxiang Xu, Gregory W. Wornell, "Lifted Residual Score Estimation". Submitted. (Note: A preliminary version of this manuscript is to be presented at *ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling*.)

[P2] **J. Jon Ryu**, Young-Han Kim, "An Information-Theoretic Proof of Kac–Bernstein Theorem," arXiv:2202.06005.

[P3] **J. Jon Ryu**, Young-Han Kim, "Minimax Optimal Algorithms with Fixed-k-Nearest Neighbors," arXiv:2202.02464v3.

Invited Talks

- **From Information Theory to Machine Learning Algorithms: Two Vignettes**.
  - Signals, Information and Algorithms Laboratory, MIT, Cambridge, MA, USA, Mar. 2022 (remote).
  - Center for AI and Natural Sciences, KIAS, Seoul, South Korea, Mar. 2022 (remote).
- **From Wyner's Common Information to Learning with Succinct Representation**.
  - Information Theory and Applications (ITA) Workshop, La Jolla, CA, USA, May 2022.
  - Machine Intelligence and Data Science Laboratory, Seoul National University, Seoul, South Korea, Jan. 2023.
  - Inference and Information for Data Science Lab, KAIST, Daejeon, South Korea, Jan. 2023.
- **Nearest Neighbor Density Functional Estimation From Inverse Laplace Transform**.
  - Center for AI and Natural Sciences, KIAS, Seoul, South Korea, Aug. 2022 (remote).
- **On Confidence Sequences from Universal Gambling**.
  - Prof. Aaditya Ramdas' Group Meeting, CMU, Pittsburgh, PA, USA, Oct. 2022 (remote).
  - Hanyang University, Seoul, South Korea, Jan. 2023.
- **Operator SVD with Neural Networks via Nested Low-Rank Approximation**.
  - MLTea talk, MIT, Cambridge, MA, USA, Nov. 2023.
  - Information Theory and Applications (ITA) Workshop, La Jolla, CA, USA, Feb. 2024.
  - Mitsubishi Electric Research Laboratories, Cambridge, MA, USA, Sep. 2024.
  - KAIST AI Graduate School of AI, Daejeon, South Korea, Sep. 2024 (remote).
  - Center for AI and Natural Sciences, KIAS, Seoul, South Korea, Oct. 2024 (remote).
  - Flatiron Institute, New York, NY, USA, Oct. 2024 (remote).
  - MIT JTL Urban Mobility Lab, Cambridge, MA, USA, Sep. 2025 (remote).
  - Prof. Devavrat Shah's Group Meeting, MIT, Cambridge, MA, USA, Nov. 2025.
- **Efficient Generative Modeling and Operator Learning from First Principles**.
  - Frontier Research, Prescient Design, Genentech, New York, NY, USA, May 2025 (remote).
  - Department of EE, POSTECH, Pohang, South Korea, Jun. 2025 (remote).
  - Department of CSE, POSTECH, Pohang, South Korea, Aug. 2025 (remote).
- **Tools for Scalable and Reliable Scientific Inference from First Principles**.
  - Prescient Design, Genentech, Sep. 2025 (remote).

Honors and Awards

Wormley Family Early Career Professorship 2026 – 2029
Penn State

Spotlight Recognitions (top ∼3% acceptance rate) ICML 2024, ICML 2025

Top Reviewer Awards NeurIPS 2023, NeurIPS 2025, ICLR 2026, ICML 2026

Departmental Fellowship Sep. 2015 – Jun. 2016
Department of ECE, UCSD

Kwanjeong Scholarship for Graduate Study Sep. 2015 – Jun. 2020
Kwanjeong Scholarship Foundation, South Korea

Kwanjeong Scholarship for Undergraduate Study Mar. 2010 – Dec. 2013
Kwanjeong Scholarship Foundation, South Korea

University Students Contest of Mathematics
Korean Mathematical Society
- Among non-math majors: Gold prize (2010), Honorable mention (2009)
- Among math majors: Bronze Prize (2013)

Reviewer Services

- **Journal**: IEEE Trans. Info. Theory, IEEE J. Selected Areas Info. Theory, J. Comput. Phys., Trans. Mach. Learn. Res.
- **Conference**: ISIT (2017, 2023, 2024, 2025), ITW 2022, AISTATS (2022, 2023, 2024), ICML (2022, 2023, 2024, 2025. 2026), NeurIPS (2022, 2023 (top 10% reviewer), 2024, 2025 (top 10% reviewer)), ICLR (2024, 2026 (top-200 reviewer), 2027).

Internship Experience

Research Intern Jun. 2019 – Dec. 2019
AI Research Group, Qualcomm Technologies, Inc.

Research Intern Jun. 2018 – Sep. 2018
Deep Learning Team, SoC R&D, Samsung Semiconductor Inc.

Teaching Experience

Instructor (Penn State)

- **A-I 410 AI Systems and Tools** Fall 2026

Instructor (MIT)

- **6.7800 Inference and Information** Spring 2024, Spring 2025
  · Designed and taught new research-related topics as special sessions as a co-instructor.
  · **Topics**: minimax bit prediction, universal inference and concentration, universal learning approach for Rock–Paper–Scissor machine, variational perspective on generative modeling.

Teaching Assistant (UCSD)

- **ECE 250 Random Processes** Winter 2017
- **ECE 154C Communication Systems** Spring 2017
  · Designed hands-on programming assignments for the class based on Julia.
  · **Topics**: Basic source coding and channel coding algrotihms.
- **ECE 225B Universal Probability and Applications in Data Science** Spring 2018
  · Designed hands-on programming assignments for the class based on Python.
  · **Topics**: Lempel–Ziv probability assignment, context-tree weighting, and universal portfolio.
- **ECE 269 Linear Algebra and Applications** Winter 2019

Mentoring Experience

Tejas Jayashankar, Ph.D. Student, MIT (now at Meta) Sep. 2022 – Jul. 2025
· Topic: Developing new generative modeling techniques.
· Outcome: Co-developed a new training technique for one-step high-quality image generators [C12].

Abhin Shah, Ph.D. Student, MIT (now at Five Rings) Sep. 2022 – Aug. 2024
· Topic: (1) Learning with fairness. (2) Learning principles for unnormalized probability models.
· Outcome: Co-authored a paper on learning under fairness constraint with uncertain attributes, presented at ISIT'24. Co-authored a paper on unifying learning principles for unnormalized probability models [C13].

Maohao Shen, Ph.D. Student, MIT (now at Thinking Machines Lab) Sep. 2022 – May 2026
· Topic: (1) Efficient uncertainty quantification algorithms. (2) Efficient LLM alignment with feedback data.
· Outcome: Co-authored a paper on uncertainty quantification for black-box models, presented at NeurIPS'24 [C11]. Developing a unified RLHF framework for LLM alignment (work in progress).

Axel Adjei, M.Eng. Student, MIT Sep. 2025 – May 2026
· Advised a master thesis on gradient-based iterative methods for spectral decomposition.
· Thesis: Gradient-Based Algorithms for Eigenvalue Decomposition: An Orbital Minimization Approach.

Pavan Yeddanapudi, Undergraduate Student, MIT Sep. 2024 – Apr. 2025
· Topic: (1) Efficient estimation of information measures for high-dimensional data [C17]. (2) Score-based techniques for representation learning (work in progress).

Samuel Zhou, Undergraduate Student, MIT Mar. 2025 – May 2025
· Topic: (1) New linear-algebraic optimization framework for representation learning [C16]. (2) New techniques for interpretable representation learning (work in progress).

Other Experience

Military Service (mandatory) Mar. 2011 – Dec. 2012
Republic of Korea Army

References

Gregory W. Wornell
Professor of EECS, MIT
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Sanjoy Dasgupta
Professor of CSE, UCSD
Ph.D. Advisor
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